

Course Notes Template — Feature Demo

Every environment and shortcut, shown with its source

Reference document, not a real course

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What this file is

This is a reference build, not a lecture entry. Each section below pairs the `verbatim` source for one template feature with what it actually renders to, so you can copy the pattern you need straight into an entry file.

Math shortcuts

Defined in `preamble.tex` so you never have to type `\mathbb{R}` by hand.

```
\R, \N, \Z, \E, \Prob \quad \Var(X), \Cov(X,Y)
```

Renders as: $\mathbb{R}$, $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{E}$, $\mathbb{P}$ $\text{Var}(X)$, $\text{Cov}(X,Y)$

Theorem-style environments

`theorem`, `lemma`, and `proposition` share one counter and use bold-header, italic-body plain style. The bracketed name is optional.

```
\begin{theorem}[Complement rule]
  For any event  $A$ ,  $\text{Prob}(A^c) = 1 - \text{Prob}(A)$ .
\end{theorem}
```

Renders as:

Theorem 1 (Complement rule). *For any event A , $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.*

```
\begin{lemma}
  Every convergent sequence is bounded.
\end{lemma}
```

```
\begin{proposition}
  The empty set is a subset of every set.
\end{proposition}
```

Renders as:

Lemma 2. *Every convergent sequence is bounded.*

Proposition 3. *The empty set is a subset of every set.*

Definition-style environments

`definition`, `example`, and `exercise` use upright body text and share the same counter sequence as the theorem-style set above.

```
\begin{definition}[Independence]
  Events  $A$  and  $B$  are independent if
   $\text{Prob}(A \cap B) = \text{Prob}(A)\text{Prob}(B)$ .
\end{definition}
```

Renders as:

Definition 4 (Independence). Events A and B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

```
\begin{example}
  Rolling a fair die:  $\text{Prob}(\{1,3,5\}) = \tfrac{1}{2}$ .
\end{example}
```

```
\begin{exercise}
  Show that  $\text{Prob}(A \cup B) = \text{Prob}(A) + \text{Prob}(B) - \text{Prob}(A \cap B)$ .
\end{exercise}
```

Renders as:

Example 5. Rolling a fair die: $\mathbb{P}(\{1,3,5\}) = \frac{1}{2}$.

Exercise 6. Show that $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Remarks and proofs

`remark` is unnumbered. `proof` comes from `amsthm` automatically and ends with the usual tombstone.

```
\begin{remark}
  Independence is a modelling assumption, not something you can see
  directly in raw data.
\end{remark}
```

```
\begin{proof}
  Immediate from  $\text{Prob}(A) + \text{Prob}(A^c) = 1$ .
\end{proof}
```

Renders as:

Remark. Independence is a modelling assumption, not something you can see directly in raw data.

Proof. Immediate from $\mathbb{P}(A) + \mathbb{P}(A^c) = 1$. □

Cross-references

Attach a `\label` to any numbered environment and refer back to it with `\ref`.

```
\begin{theorem}[Bayes' rule]\label{thm:bayes}
  $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$.
\end{theorem}
```

See Theorem~\ref{thm:bayes} above.

Renders as:

Theorem 7 (Bayes' rule). $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$.

See Theorem 7 above.

Handwritten-note formatting helpers

These extra commands (also in `preamble.tex`) reproduce the look of red definition labels, blue margin comments, and yellow highlights from scanned handwritten notes; see `STA347/entries/week01.tex` for a full worked example.

```
\deflabel{2.1}{Conditional Probability}
\begin{quote}
  $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B) / \mathbb{P}(B)$ when $\mathbb{P}(B) > 0$.
\end{quote}
\aside{This is undefined if $B$ has probability zero.}
```

```
\heading{Key difference:}
```

Conditioning rescales the sample space; independence does not.

Uniform convergence needs `\hly{a single threshold $N$}` for every `$x$`.

Renders as:

Def 2.1 Conditional Probability

$$\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B) / \mathbb{P}(B) \text{ when } \mathbb{P}(B) > 0.$$

This is undefined if B has probability zero.

Key difference:

Conditioning rescales the sample space; independence does not.

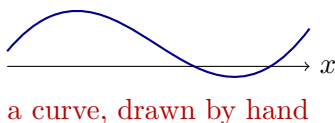
Uniform convergence needs `asinglethresholdN` for every x .

Diagrams with TikZ

`tikz` and the `decorations.pathreplacing` library are loaded by the preamble, so any entry can draw a quick sketch inline.

```
\begin{center}
\begin{tikzpicture}
  \draw[->] (0,0) -- (4,0) node[right]{$x$};
  \draw[thick, blue!55!black] (0,0.2) .. controls (1.5,2) and (2.5,-1.5)
    .. (4,0.5);
  \node[red!70!black] at (2,-0.6) {a curve, drawn by hand};
\end{tikzpicture}
\end{center}
```

Renders as:



Lists

`enumitem` is loaded with `nosep`, so list items sit close together the way handwritten bullet points do.

```
\begin{itemize}
  \item Sample space  $\Omega$ 
  \item Event space  $\mathcal{F}$ 
  \item Probability measure  $\mathbb{P}$ 
\end{itemize}
```

Renders as:

- Sample space Ω
- Event space $\mathcal{F}$
- Probability measure $\mathbb{P}$

Tables

`booktabs` gives the clean horizontal-rule table style (no vertical rules).

```
\begin{center}
\begin{tabular}{lcc}
\toprule
Distribution & Mean & Variance \\
\midrule
 $\text{Bernoulli}(p)$  &  $p$  &  $p(1-p)$  \\
 $\text{Poisson}(\lambda)$  &  $\lambda$  &  $\lambda$  \\
\bottomrule
\end{tabular}
\end{center}
```

Renders as:

Distribution	Mean	Variance
$\text{Bernoulli}(p)$	p	$p(1-p)$
$\text{Poisson}(\lambda)$	λ	λ

Links

`hyperref` is loaded with colored links, so both cross-references (Section) and external URLs render in color automatically.

See `\href{https://www.overleaf.com}{Overleaf}` for an online editor.

Renders as: See [Overleaf](https://www.overleaf.com) for an online editor.

Images

`graphicx` is loaded but no image ships with this demo, so the call below is shown only as source — point it at a real file in a course-local `figures/` folder to use it.

`\includegraphics[width=0.6\textwidth]{figures/diagram.png}`